Please check the examination details below before entering your candidate information

Candidate surname	Other names

Group

Candidate Number

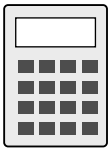
Elite IGCSE Academy | International GCSE

Expected Paper

November 2026

Time: **2 hours**

Paper reference **ELT/U2X**



Calculator allowed

Mathematics A

MODULAR · UNIT 2

Higher Tier · Expected Exam

You must have: Ruler graduated in centimetres and millimetres, protractor, pair of compasses, pen, HB pencil, eraser, calculator.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, group and candidate number.
- Answer **all** questions.
- Without sufficient working, correct answers may be awarded no marks.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- **Calculators may be used.**
- You must **NOT** write anything on the formulae page.

Information

- The total mark for this paper is 100.
- There are **26** questions in this paper.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Questions 19 to 26 are the most demanding. Aim to reach them with at least 45 minutes left. Aim to reach them with at least 45 minutes left.
- Check your answers if you have time at the end.

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Dr Eslam Ahmed
Tel. **01120009622**
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ELITE IGCSE ACADEMY

ELT U2X 2026

International GCSE Mathematics

Formulae sheet – Higher Tier

Arithmetic series

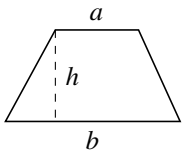
Sum to n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$

The quadratic equation

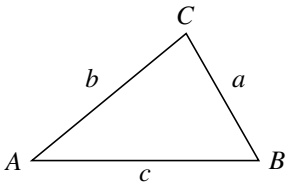
The solutions of $ax^2 + bx + c = 0$ where $a \neq 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Area of trapezium $= \frac{1}{2}(a+b)h$



Trigonometry



In any triangle ABC

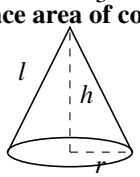
Sine Rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine Rule $a^2 = b^2 + c^2 - 2bc\cos A$

Area of triangle $= \frac{1}{2}ab\sin C$

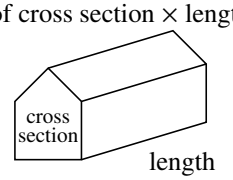
Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$



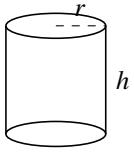
Volume of prism

= area of cross section × length



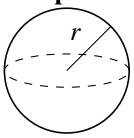
Volume of cylinder $= \pi r^2 h$

Curved surface area of cylinder $= 2\pi r h$



Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$



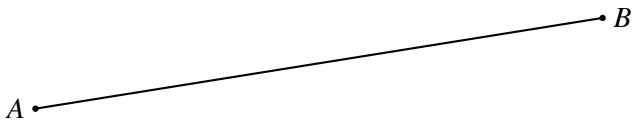
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Answer ALL TWENTY SIX questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 Use ruler and compasses only to construct the perpendicular bisector of the line AB.
You must show all your construction lines.



(Total for Question 1 is 2 marks)

2 The table gives information about the heights, h cm, of 40 plants.

Height (h cm)	**Frequency**
$0 < h \leqslant 20$	6
$20 < h \leqslant 40$	14
$40 < h \leqslant 60$	18
$60 < h \leqslant 80$	2

(a) Write down the modal class.

. . . . . .

(1)

(b) Work out an estimate for the mean height of the 40 plants.

. . . . . cm

(3)

(Total for Question 2 is 4 marks)

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3 Omar buys 40 boxes of cakes for \$12 each.
He sells 25 of the boxes for \$20 each and the rest of the boxes for \$15 each.
Work out Omar's percentage profit. Give your answer correct to 3 significant figures.

. . . . . %

(3)

(Total for Question 3 is 3 marks)

---

4 Some money is shared between Ali, Bilal and Celine in the ratio 2 : 3 : 5
Celine gets \$84 more than Ali.
Work out the total amount of money that is shared.

\$ . . . . . .

(3)

(Total for Question 4 is 3 marks)

5

(a) Write 0.000 607 in standard form.

. . . . . .

(1)

(b) Work out $\dfrac{5.4 \times 10^{9}}{1.2 \times 10^{-4}}$ Give your answer in standard form.

. . . . . .

(2)

(Total for Question 5 is 3 marks)

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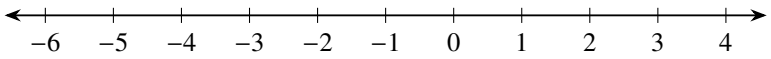
6

(a) Solve the inequality $5 - 3x \leqslant 2x + 20$

. . . . . .

(2)

(b) Show your answer to part (a) on the number line.



(1)

(Total for Question 6 is 3 marks)

---

7 In a sale, the price of a television is reduced by 30%.
The sale price of the television is $434
Work out the price of the television before the sale.

$ . . . . . .

(3)

(Total for Question 7 is 3 marks)

---

8 The size of each interior angle of a regular polygon is 4 times the size of each exterior angle.
Work out the number of sides of the polygon.

. . . . . .

(3)

(Total for Question 8 is 3 marks)

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9 Kai invests \$6500 for 5 years at 2.8% per year compound interest.

Work out the value of his investment at the end of 5 years. Give your answer to the nearest cent.

\$ .....

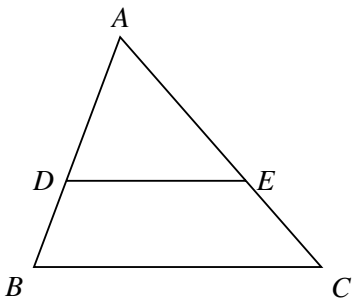
(3)

(Total for Question 9 is 3 marks)

---

10 ADB and AEC are straight lines.
DE is parallel to BC.
$AD = 5\,\text{cm}$ $DB = 3\,\text{cm}$ $DE = 6.5\,\text{cm}$ $AE = 4.5\,\text{cm}$

Diagram **NOT** accurately drawn



(a) Work out the length of BC.

..... cm

(2)

(b) Work out the length of EC.

..... cm

(2)

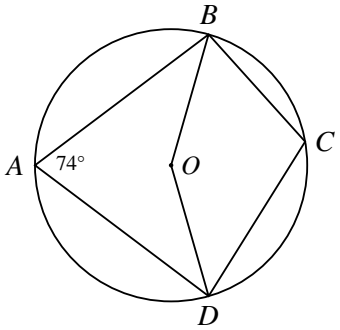
(Total for Question 10 is 4 marks)

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11 A, B, C and D are points on a circle, centre O.
Angle $BAD = 74°$

Diagram **NOT** accurately drawn



(a) Work out the size of angle BCD.
Give a reason for your answer.

. . . . . °

(2)

(b) Work out the size of the obtuse angle BOD. Give a reason for your answer.

. . . . . °

(2)

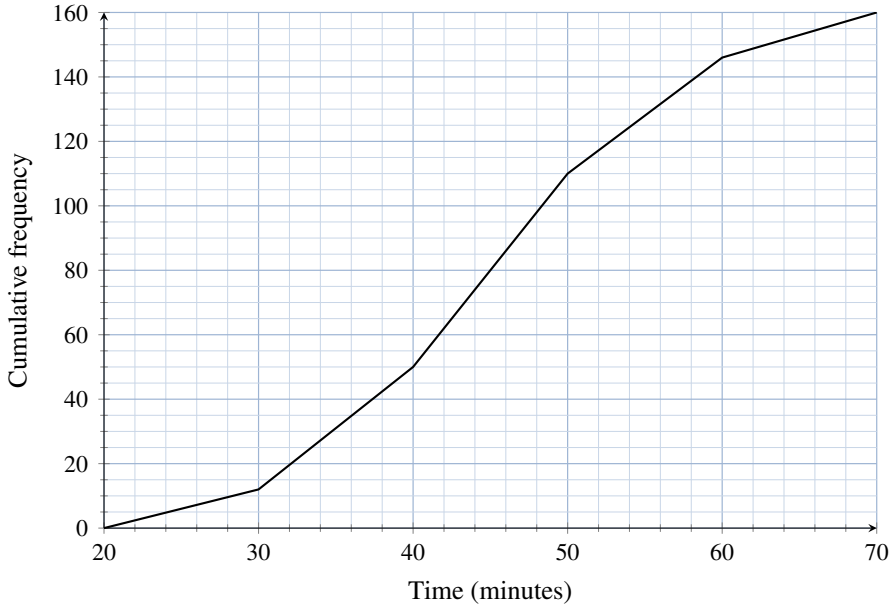
(Total for Question 11 is 4 marks)

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12 The cumulative frequency graph shows the times, in minutes, that 160 runners took to finish a race.



(a) Use the graph to find an estimate for the median time.

. . . . . minutes

(1)

(b) Use the graph to find an estimate for the interquartile range of the times.

. . . . . minutes

(2)

(c) Use the graph to find an estimate for the number of runners who took more than 55 minutes.

. . . . . .

(2)

(Total for Question 12 is 5 marks)

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13 F is inversely proportional to the square of d.
When $d = 3$, $F = 20$

(a) Find a formula for F in terms of d.

$F =$ .....

(2)

(b) Work out the value of F when $d = 6$

$F =$ .....

(1)

(c) Work out the positive value of d when $F = 0.45$

$d =$ .....

(1)

(Total for Question 13 is 4 marks)

---

14 The volume of a sphere is 288π cm^3
Work out the surface area of the sphere. Give your answer in terms of π.

..... cm^2

(3)

(Total for Question 14 is 3 marks)

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15 Make k the subject of the formula $p = \dfrac{8k + 5}{3k - 2}$

$k =$ .....

(4)

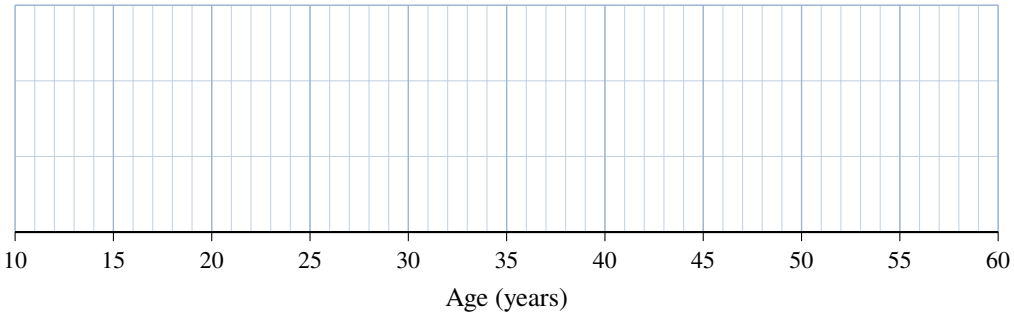
(Total for Question 15 is 4 marks)

---

16 The information below is about the ages of the people in a club.

The youngest person is 18 years old and the oldest is 52 years old.
The median age is 31 years. The lower quartile is 24 years. The interquartile range is 15 years.

On the grid, draw a box plot for this information.



(3)

(Total for Question 16 is 3 marks)

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17 $f(x) = 3 - 2x$ $\quad g(x) = \dfrac{4}{x+1}$

(a) Find $fg(3)$

.....

(1)

(b) State which value of x must be excluded from any domain of g.

.....

(1)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \ldots$

$g^{-1}(x) =$ .....

(2)

(Total for Question 17 is 4 marks)

---

18 OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$
M is the midpoint of AB.

(a) Find $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....

(1)

(b) Find $\overrightarrow{OM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. Give your answer in its simplest form.

.....

(2)

(Total for Question 18 is 3 marks)

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19 The curve C has equation $y = x^3 - 4x^2 + 5$
A is the point on C with x-coordinate 2

(a) Work out the gradient of C at A.

.....

(2)

(b) There is one other point on C where the gradient is the same as the gradient at A.
Find the coordinates of this point. Show clear algebraic working.

(..... , ...)

(3)

(Total for Question 19 is 5 marks)

---

20 The line with equation $y = x + k$ is a tangent to the circle with equation $x^2 + y^2 = 8$

Find the two possible values of k.
Show clear algebraic working.

.....

(5)

(Total for Question 20 is 5 marks)

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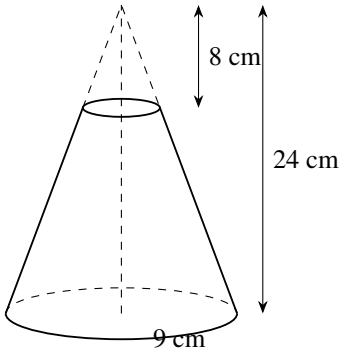
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21 A solid cone has base radius 9 cm and height 24 cm.
A smaller cone of height 8 cm is cut off the top of the cone, leaving a frustum.

The frustum is melted down and all the metal is made into a solid sphere.

Work out the radius of the sphere.
Give your answer correct to 3 significant figures.

Diagram **NOT** accurately drawn



. . . . . cm

(5)

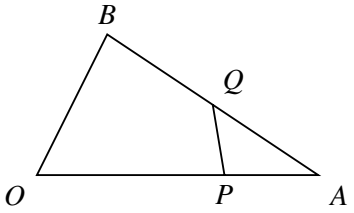
(Total for Question 21 is 5 marks)

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22

OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$
P is the point on OA such that $OP : PA = 2 : 1$
Q is the midpoint of AB.

Diagram **NOT** accurately drawn



(a) Find $\overrightarrow{PQ}$ in terms of **a** and **b**. Give your answer in its simplest form.

. . . . . .

(2)

The line PQ is extended to meet the line OB extended at the point R.

(b) Using a vector method, find $\overrightarrow{OR}$ in terms of **b**.

. . . . . .

(3)

(Total for Question 22 is 5 marks)

23 Here are the first four terms of an arithmetic series.

7 11 15 19

Find the least number of terms of the series that must be added together so that the sum is greater than 1000
Show clear algebraic working.

. . . . . .

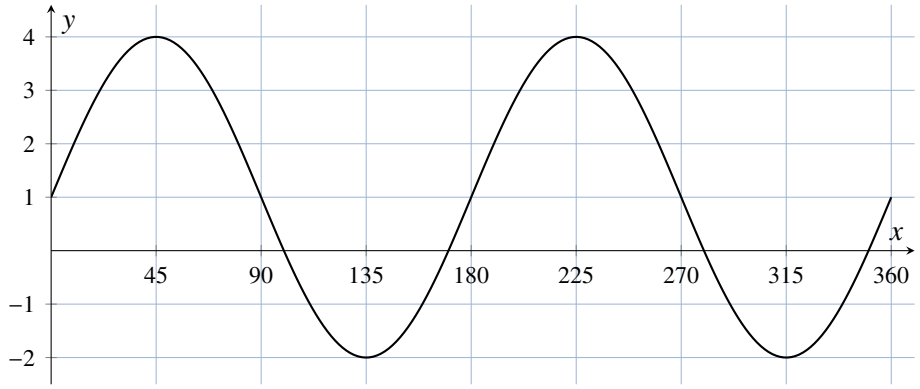
(4)

(Total for Question 23 is 4 marks)

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24 The graph of $y = a\sin(bx^\circ) + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid.



(a) Find the value of a, the value of b and the value of c.

$a =$ .....

$b =$ .....

$c =$ .....

(3)

(b) Write down the number of solutions of $a\sin(bx^\circ) + c = 2.5$ for $0 \leqslant x \leqslant 360$

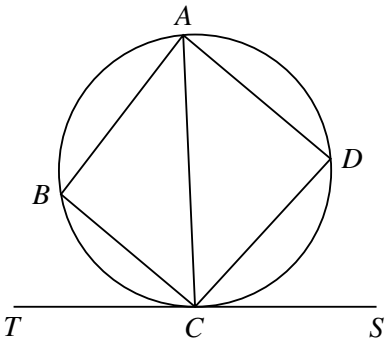
.....

(1)

(Total for Question 24 is 4 marks)

25 A, B, C and D are points on a circle.
TCS is the tangent to the circle at C.

Diagram **NOT** accurately drawn



Angle $ABC = (3x + 10)°$
Angle $ADC = (2x + 20)°$
Angle $DCS = (x + 15)°$

Work out the size of angle ACD.
Give a reason for each stage of your working.

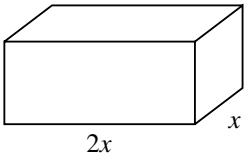
. . . . . °

(5)

(Total for Question 25 is 5 marks)

26 A closed box is in the shape of a cuboid.
The base of the box is a rectangle of length $2x$ cm and width x cm.
The volume of the box is $72\,\text{cm}^3$
The total surface area of the box is $A\,\text{cm}^2$

Diagram **NOT** accurately drawn



(a) Show that $A = 4x^2 + \frac{216}{x}$

(3)

(b) Use calculus to find the minimum value of A.
Show your working clearly.

. . . . . .

(3)

(Total for Question 26 is 6 marks)

TOTAL FOR PAPER IS 100 MARKS